

Effect of fruit restriction on glycemic control in patients with type 2 diabetes – a randomized trial



Background Medical nutrition therapy is recognized as an important treatment option in type 2 diabetes. Most guidelines recommend eating a diet with a high intake of fiber-rich food including fruit. This is based on the many positive effects of fruit on human health. However some health professionals have concerns that fruit intake has a negative impact on glycemic control and therefore recommend restricting the fruit intake. We found no studies addressing this important clinical question. The objective was to investigate whether an advice to reduce the intake of fruit to patients with type 2 diabetes affects HbA1c, bodyweight, waist circumference and fruit intake. Methods This was an open randomized controlled trial with two parallel groups. The primary outcome was a change in HbA1c during 12 weeks of intervention. Participants were randomized to one of two interventions; medical nutrition therapy + advice to consume at least two pieces of fruit a day (high-fruit) or medical nutrition therapy + advice to consume no more than two pieces of fruit a day (low-fruit). All participants had two consultations with a registered dietitian. Fruit intake was self-reported using 3-day fruit records and dietary recalls. All assessments were made by the “intention to treat” principle. Results The study population consisted of 63 men and women with newly diagnosed type 2 diabetes. All patients completed the trial. The high-fruit group increased fruit intake with 125 grams (CI 95%; 78 to 172) and the low-fruit group reduced intake with 51 grams (CI 95%; -18 to -83). HbA1c decreased in both groups with no difference between the groups (diff.: 0.19%, CI 95%; -0.23 to 0.62). Both groups reduced body weight and waist circumference, however there was no difference between the groups. Conclusions A recommendation to reduce fruit intake as part of standard medical nutrition therapy in overweight patients with newly diagnosed type 2 diabetes resulted in eating less fruit. It had however no effect on HbA1c, weight loss or waist circumference. We recommend that the intake of fruit should not be restricted in patients with type 2 diabetes. Trial registration <http://www.clinicaltrials.gov>; Identifier: NCT01010594.